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Review article

Hyperscanning neuroimaging technique to reveal the "two-in-one" system in social interactions

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ABSTRACT

Using a technique for measuring brain activity simultaneously from two people, known as hyperscanning, we can calculate inter-brain neural effects that appear only in interactions between individuals. Hyperscanning studies using fMRI are advantageous in that they can precisely determine the region(s) involved in inter-brain effects. However, it is almost impossible to record inter-brain effects in daily life. By contrast, hyperscanning EEG studies have high temporal resolution and could be used to capture moment-to-moment interactions. In addition, EEG instrumentation is portable and easy to wear, offering the opportunity to record inter-brain effects during daily-life interactions. However, the disadvantage of this approach is that it is difficult to localize the epicenter of the inter-brain effect. fNIRS has better temporal resolution and portability than fMRI, but has limited spatial resolution and a limited ability to record deep brain structures. Future studies should employ hyperscanning EEG–fMRI, because this approach combines the high temporal resolution of EEG with the high spatial resolution of fMRI. Hyperscanning EEG–fMRI allows us to use inter-brain effects as neuromarkers of the properties of social interactions in daily life. We also wish to emphasize the need to develop a mathematical model explaining how two brains can exhibit synchronized activity.

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1. Introduction

Social communication skills are essential for humans. One of the reasons why humans were able to build a highly organized civilization is that we have developed social communication skills, e.g., language. Conventionally, the neural basis of social communication has been explored through studies of brain lesions and patients with communication disorders (for review, Frith and Frith, 2001; Wood and Grafman, 2003). In addition, the combination of neuroimaging techniques and psychological experiments performed on normal subjects engaging in social behaviors has clarified the neural basis of communication (Gallotti and Frith, 2013). One of the limitations of conventional studies, however, is that they have mainly focused on aspects of off-line social cognition, whereas most of our social behavior is characterized by on-line mutual interaction, forming a "two-in-one" system

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(Konvalinka and Roepstorff, 2012; Schilbach et al., 2013). The two-in-one system in social communication is a complex nonlinear system (Beer, 2000; Froese et al., 2013) that cannot be reduced to the summation of effects in single isolated brains (Hari and Kujala, 2009; Konvalinka and Roepstorff, 2012). Therefore, it seems logical to simultaneously record two-brain activity during social interactions.

As far as we know, Duane and Behrendt (1965) were the first to simultaneously record multi-subject brain activity using the electroencephalogram (EEG). They attempted to consider two brains as a single two-in-one system by calculating the correlation between EEG traces acquired from two individuals. Although the multi-subject recording technique was subsequently ignored for a long period of time, the idea was resurrected in this century and renamed the "hyperscanning technique" (Montague et al., 2002). Since then, several researchers have employed hyperscanning to reveal how two brains interact with each other during social interactions (for review, Astolfi et al., 2011; Sänger et al., 2011; Dumas, 2011; Babiloni and Astolfi, 2012; Hari et al., 2013; Hasson et al., 2012; Scholkmann et al., 2013; Konvalinka and Roepstorff, 2012; Schilbach et al., 2013).

The term "hyperscanning" refers to simultaneous recording of brain activation from multiple subjects. Some hyperscanning studies have only analyzed how single brains are activated during social interaction (Krill and Platek, 2012; Fliessbach et al., 2012; Morita et al., 2014; Smith et al., 2014). Social communication emerges as two individuals act upon each other, and this communication constitutes a two-in-one nonlinear complex via inter-subject correlation of behavior and neural activation (Beer, 2000; Hari and Kujala, 2009; Froese et al., 2013). Therefore, for the purpose of exploring the properties of social interactions, the most efficient way to use hyperscanning neuroimaging data is to calculate inter-brain effects such as correlative (functional connectivity) or causal (effective connectivity) relationships across regions within the two brains. Several studies have introduced inter-subject correlation analysis, which calculates voxel-wise temporal correlations using Pearson's correlation coefficient between every pair of subjects who watched the same movie (Hasson et al., 2004; Englander et al., 2012; Salmi et al., 2013) and a series of faces (Burgess, 2013). We will not focus on those studies here, however, because the experiments involved no on-line social interactions between pairs of individuals.

In this review, we will briefly examine recent studies of inter-brain neural synchronization during social interactions, and also discuss the limitations of these studies. Following that, we propose future directions for hyperscanning neuroimaging: hyperscanning EEG–fMRI and mathematical modeling of inter-brain effects.

2. Hyperscanning fMRI

Functional magnetic resonance imaging (fMRI) is capable of precisely defining the epicenter of brain activity. The spatial resolution of fMRI (2–3 mm) is significantly greater than that of electroencephalogram (EEG; 1–2 cm) (Kaiboriboon et al., 2012), or functional near-infrared spectroscopy (fNIRS; 2–3 cm) (Huettel et al., 2009). In addition, only fMRI can record activation of deep brain structures. The regions associated with our social skills are located in deep brain structures such as the medial prefrontal cortex, orbitofrontal cortex, striatum, and amygdala (Allison et al., 2000; Gallotti and Frith, 2013). In general, scalp EEG and fNIRS cannot record such deep structures. Although the epicenter can be estimated mathematically using EEG (Pascual-Marqui et al., 1994), precise definition is difficult when there are multiple epicenters, or the epicenters are distant from the electrodes (Grech et al., 2008). The same is true in hyperscanning neuroimaging (Hari et al., 2013). These lines of evidence suggest that fMRI should be regarded as the first-line technique for determining which brain regions constitute the inter-brain neural network during social interactions.

The size of MRI scanners becomes a weakness when we consider using hyperscanning fMRI to reveal the nature of social interactions; however, this issue is surmountable, at least to some extent. One might think that we would need two or more MRI scanners in one institute to conduct such an experiment. However, this is not necessary: in a previous hyperscanning study (King-Casas et al., 2005), one subject was placed in a scanner in Texas, and the other was in a scanner at California. Synchronization was achieved through broadband Internet connections among multiple computers with a timing precision of approximately ±50 ms (King-Casas et al., 2005). A recent study reported that synchronization via Internet connections is sufficient for magnetoencephalography (MEG) systems, which require more accurate synchronization (Baess et al., 2012). Thus, by connecting two scanners at different locations, we can realize a hyperscanning fMRI system even though the scanners are not present in the same institute. Nonetheless, the MRI scanner limits the natural movement of participants in experiments so that it is almost impossible to record the brain during everyday social interaction. We expect that combining EEG and fMRI could resolve this limitation. The issue is addressed in more detail in a later chapter.

So far, only few hyperscanning fMRI studies have attempted to evaluate inter-brain neural synchronization during on-line social interactions (Scholkmann et al., 2013). Saito et al. (2010) investigated how two brains could be connected via eye contact. In their experiment, participants were paired with a partner of the same gender. Each subject was positioned in the MR scanner to allow collection of brain activation as they performed a joint attention (JA) task. In the scanner, they were able to see each other's eyes through the mirror using MR-compatible video camera systems, so that they could communicate on-line via eye contact. We should note that eye contact was maintained at baseline while the subjects engaged in on-line gaze exchange in the JA task. To test whether inter-brain neural synchronization was caused by eye contact rather than the JA task itself, the authors modeled out all the JA-related effects using a general linear model or autoregressive models (Fair et al., 2007). After that, they calculated inter-brain synchronization between voxels encoding homologous locations. They found that paired subjects exhibited more prominent inter-brain neural synchronization than non-paired subjects in the right IFG (Brodmann area (BA) 44/45/47), and concluded that the IFG is involved in sharing intention through eye contact, which provides the context for JA. This conclusion was supported by the lack of inter-brain neural synchronization in people with autistic spectrum disorder (ASD), who experience difficulty in sharing intention through eye contact (Tanabe et al., 2012).

fMRI scanning causes very intense acoustic noise, making verbal communication during an experiment difficult. However, using a noise-canceling fiber-optic microphone system, a recent preliminary study showed that inter-brain neural synchronization occurs between the speech production and auditory areas during conversation (Spiegelhalder et al., 2014). By introducing a more sophisticated paradigm, we could use hyperscanning fMRI to elucidate how two brains engage in inter-brain neural synchronization during conversation.

3. Hyperscanning EEG

EEG is the most frequently used technique in hyperscanning studies (Scholkmann et al., 2013). One advantage of EEG is that it has finer temporal resolution than fMRI. Although a recently developed high-speed recording technique enables recording of

fMRI data on the sub-second timescale (Moeller et al., 2010), EEG offers the opportunity to record activation on the millisecond scale (Huettel et al., 2009). This superior temporal resolution could assist in estimating the causal relationships between brain activation acquired from two individuals (Konvalinka and Roepstorff, 2012). The fine temporal resolution provides the opportunity to assess frequency dependency in inter-brain neural synchronization, and to investigate instantaneous inter-brain synchronization. Another advantage of EEG is that the instrumentation is portable and easy to wear; consequently, EEG recording devices do not limit the activities in which participants can engage. This feature could enable us to discuss inter-brain neural synchronization in more natural settings.

The biggest disadvantage of EEG is its limited ability to localize the epicenter of brain activation (Huettel et al., 2009). EEG measures the scalp electric potential generated by neuronal currents near the brain surface, and its spatial resolution is in the range of 1–2 cm because of the inhomogeneous conductivity profile of the head (Kaiboriboon et al., 2012). By contrast, the fMRI has a spatial resolution of 2–3 mm (Huettel et al., 2009). Although mathematical methods allow us to estimate the epicenter of EEG activity (Pascual-Marqui et al., 1994), source estimation is an ill-posed problem (Grech et al., 2008), and it is almost impossible to define the source points when they are distant from the scalp electrodes. Consequently, scalp EEG cannot record social regions located in deep brain structures (Allison et al., 2000; Gallagher and Frith, 2003). For example, Yun et al. (2012) reported an inter-brain network involving the parahippocampal gyrus (PHG), but it was suspicious that an EEG source in a deep structure such as the PHG could be precisely localized (Hari et al., 2013). Because of these difficulties in localizing the epicenter of activation, we have to note that EEG seems inappropriate for precisely determining the spatial pattern of the inter-brain networks involved in social interactions.

To the best of our knowledge, Dumas et al. (2010) were the first to investigate inter-brain neural coupling during face-to-face social communication using a hyperscanning EEG system. In their communication task, the 'imitator' had to spontaneously imitate the hand action of the 'model'. The authors found that the alpha–mu (8–12 Hz) band exhibited the most robust inter-brain EEG synchronization in the right centro-parietal cortex. Interestingly, another EEG study reported that the alpha–mu band in the right centro-parietal region was a significant neuromarker of social coordination (Tognoli et al., 2007), and that this region mediates how a participant interprets their partner's body movement in a social context (Naeem et al., 2012). Based on these lines of evidence, the authors concluded that several aspects of ongoing imitation tasks, such as anticipation of another's actions and co-regulation of turn-taking, result in inter-brain EEG synchronization.

Both verbal and nonverbal communications are necessary in order for two people to play a game together. Using a card game and hyperscanning EEG, Astolfi et al. (2010) revealed the association between the cooperative relationship with a partner and inter-brain neural synchronization between partners. In their study, four participants were divided into two teams, and these two teams competed against each other in the game. An extension of the Granger-causality approach was used to evaluate the causal relationship between any two participants, i.e., either a pair belonging to the same team, or a pair belonging to different teams. The EEG activity corresponding to the prefrontal regions (BA 8/9/46) had an effect on activity in the anterior cingulate cortex (ACC) in the partner's brain, but only when the two participants belonged to the same team. The frequencies exhibiting this causal relationship were in the alpha (8–12 Hz), beta (13–29 Hz), and gamma (30–40 Hz) bands. The authors speculated that this synchronization reflects the cooperative relationship between team members (Astolfi et al., 2010).

Another study showed that the psychological perspective required for cooperation with a partner is represented by inter-brain EEG synchronization as two subjects play the prisoner's dilemma game (De Vico Fallani et al., 2010). In that study, brain activity was recorded by hyperscanning EEG, and the non-linear classifier method was introduced to reveal whether a player's strategy in the game was represented by an inter-brain network pattern. When both participants intended to defect on their partners, there was less inter-brain neural synchronization than in pairs that had chosen cooperative strategies. The characteristics of the inter-brain network prior to decision-making could predict the strategies that were actually chosen with up to 90% accuracy. Therefore, the authors indicated that inter-brain synchronization might be a marker that can predict relationships in social interaction.

A series of studies by Lindenberger and colleagues, which focused on subjects cooperatively playing the guitar in a duet, clearly indicated that inter-brain neural effects arise even in daily life. First, the authors found that the enhancement of an inter-brain effect, i.e. inter-brain EEG phase coherence between the central and frontal nodes, is conspicuous during periods that require particularly high demands on musical coordination (Sänger et al., 2012). This observation supports the idea that the inter-brain network plays a functional role in coordination of behavior. Second, graph-based analysis revealed that the inter-brain network during guitar playing exhibits "small world" properties, as with intra-brain functionally connected networks (Sänger et al., 2012). Small-worldness means that the network acts as functionally integrated network (Sporns and Zwi, 2004), suggesting that inter-brain networks play a functional role in social interactions. Third, the characteristics of inter- or intra-brain networks clearly reflected the musical roles during improvising together. The inter- and intra-brain network associated with playing guitar in a duet differs significantly from that involved in listening to a partner play the guitar (Müller et al., 2013). In addition, even during duet playing, the phase locking between subjects' brains was modulated in relation to their experimentally assigned musical roles, i.e., leader or follower (Sänger et al., 2013). Based on these findings, the authors speculated that the strength and spatio-temporal pattern of inter-brain effects are associated with the ability to coordinate one's own behavior with that of a partner.

Yun et al. (2012) reported that the inter-brain EEG effect can be modulated by training. In pre-training sessions, participants were instructed to look at their partner's index finger while holding their own arms extended and keeping their own index fingers together and as stationary as possible. Both inter-brain EEG synchronization and unconscious arm movements were recorded. In the training session following the pre-training sessions, paired participants were assigned roles as leader and follower. The leader randomly moved his or her index finger while keeping his or her arm extended, and the follower followed it. Participants learned the leader–follower relationship through training. After the training, coupled participants repeated a post-training session that was identical to the pre-training sessions. Inter-brain EEG synchronization increased after training, specifically, from the leader's PHG, anterior cingulate (ACC), and postcentral gyrus (PoCG), to the follower's inferior frontal gyrus (IFG) in the theta (4–7.5 Hz) and beta (12–30 Hz) frequency bands. Based on these results, the authors concluded that the emergence of inter-brain EEG synchronization is not merely stimulus-driven, but can be modulated by training.

The training effect was also mentioned in Kawasaki et al. (2013). In that experiment, two participants engaged in an alternating speech task in which they pronounced letters one after the other. For example, one subject said 'A', then the other said 'B', and then the procedure was repeated. First, participants performed the task with a human (human pre-machine condition). In that case, the amplitude of EEG waveforms in the theta/alpha (6–12 Hz)

frequency band exhibited significant inter-brain correlation between the homologous temporal and lateral–parietal electrodes. After the human condition, participants performed the same task with a robot-like machine (machine condition) as training. After that, they repeated the same task with a human (human post-machine condition). Inter-brain neural and speech synchronization was clearly enhanced in the post-machine condition relative to the pre-machine condition. The authors claimed that this enhancement reflected the emergence of empathy for others' behavior.

4. Hyperscanning fNIRS

fNIRS is less capable than fMRI of localizing the epicenter of activation. Because of the underlying principles of the measurement, fNIRS can only record the brain surface, and the spatial resolution is under 1 cm (Huettel et al., 2009). In addition, fNIRS has high sensitivity to blood-flow changes in the scalp that are unrelated to brain activation (Gregg et al., 2010; Kirilina et al., 2012). These features make it difficult for fNIRS to acquire pure neural activation. However, because fNIRS is much more portable than fMRI, the number of studies using this technique to study inter-brain effects has been growing rapidly (Scholkmann et al., 2013).

To the best of our knowledge, the first hyperscanning fNIRS study was performed by Funane et al. (2011). In that study, participants sat down face-to-face, and were requested to synchronously press a button after counting 10 s in their mind. The relationship between task performance and inter-brain neural coherence was assessed using fNIRS, which was simultaneously recorded from both participants. Higher inter-brain coherence in the prefrontal cortex was significantly associated with shorter time intervals between button presses. The authors concluded that performance of cooperative behavior is associated with inter-brain coherence.

Cui and colleagues also assessed whether inter-brain effects influence cooperative task performance (Cui et al., 2012). In their experiment, participants were required to press a button as fast as possible after a visual cue signal appeared on the screen. There were two types of goal structures for the task: cooperative and competitive. In the cooperative task, when the average response time was shorter than the previously defined threshold, participants could win points in the game. In the competitive task, participants had to press a button before their counterpart did. Participants had to repeat each task twice. Wavelet transformation coherence was used to evaluate inter-brain effects. The percentage of winning trials increased in the second cooperative task block relative to first block, whereas there were no increases in competitive tasks. Additionally, the repetition of the task increased inter-brain coherence in the right superior frontal cortices. Given these findings, the authors concluded that inter-brain coherence might be a proxy for performance of cooperative tasks (Cui et al., 2012).

Using hyperscanning fNIRS, it has reported that the inter-brain coherence represents the effort to imitate partner's movement (Holper et al., 2012). In the study, two distinct roles were assigned to participants: the model and imitator. The model was asked to perform a paced finger tapping task by repeatedly pressing a button on a keyboard using right hand fingers (index, middle, ring, pinky and thumb). The order of finger used could be varied by the model. Another participant, i.e. imitator, was required to imitate model's finger tapping. In the control condition, participants asked to perform the finger tapping task alone. The authors reported the increased inter-brain coherence during imitating task as compared to control condition in the left frontal area. In addition, enhanced coherence was conspicuous when imitator maintains the button press in self pace, as compared when participants were asked to model the movement in synchrony with a metronome. The authors confirmed that the increased inter-brain coherence could not be accounted by increased behavioral synchrony. As with the hyperscanning EEG study (Sänger et al., 2012), this hyperscanning fNIRS study supports the idea that the similarity between movement patterns did not contribute to the inter-brain coherence during imitation, but that the coherence reflected the effort of imitation.

Another study showed that conversation can cause inter-brain coherence (Jiang et al., 2012). In that experiment, participants talked about two hot news topics using different communication styles: face-to-face dialog, back-to-back dialog, face-to-face monolog, and back-to-back monolog. Only in the face-to-face dialog condition, there was significant inter-brain synchronization in the probe corresponding to the left inferior frontal cortex (IFC). In addition, the strength of IFC synchronization could accurately predict the occurrence of nonverbal communication behaviors such as turn-taking and body language. These findings suggest that the IFC synchronization is mediated by non-verbal behaviors (Jiang et al., 2012).

Inter-brain effects exhibit different patterns when verbal information is absent in vocal communication. Using hyperscanning fNIRS, Osaka et al. (2014) also attempted to reveal how eye-to-eye communication could link two brains during a cooperative humming task. In the eye-to-eye condition, paired participants cooperatively hummed a song while watching each other's eyes. In the non-eye-to-eye condition, participants watched a wall placed between them. Wavelet transform coherence was used as the index of the inter-brain effect, and the authors compared the coherence in eye-to-eye condition with that in the non-eye-to-eye condition. Inter-brain coherence in the right IFC was conspicuous only during non-eye-to-eye humming. This result suggests that inter-brain coherence in the right hemisphere is caused by non-verbal coordination such as humming (Osaka et al., 2014), whereas synchronization in the left hemisphere results from verbal dialog (Jiang et al., 2012).

Dommer et al. (2012) attempted to reveal the origin of inter-brain neural synchronization using hyperscanning fNIRS as subjects played the dual n-back task. There were two settings: either participants were paired and sat side-by-side performing the n-back task in a joint manner (paired condition), or they performed the task alone (single condition). The inter-brain effect was calculated using wavelet transform coherence. The study revealed that inter-brain coherence significantly increased in the prefrontal cortex during the paired condition relative to the single condition, suggesting that coordination of behavior binds two brains together. In addition, there two frequency bands contributed to the inter-brain neural synchronization: the heart rate frequency of each participant and low frequency (approximately 0.1 Hz). In a single-brain resting-state network study, it was confirmed that the functionally connected network in the low-frequency range reflects spontaneous neural activities, whereas the higher frequencies are contaminated by heart beating and respiration (Cordes et al., 2001). Based on these findings, the authors concluded that the low-frequency inter-brain network represents coordinated behaviors performed in a joint manner (Dommer et al., 2012).

5. Necessity of hyperscanning EEG–fMRI

It is assumed that inter-brain effects are tightly associated with properties of interaction such as task structure, interaction structure, and goal structure (Schilbach et al., 2013; Liu and Pelowski, 2014). Hyperscanning EEG study has already revealed that goal structure (i.e., competitive vs. cooperative task) has an effect on inter-brain network patterns (Astolfi et al., 2010; Cui et al., 2012; Funane et al., 2011). The social interactions that occur in real life exhibit more complex patterns than those that arise in experiments. At present, it is not obvious whether there is an inter-brain

effect reflecting the nature of interactions, even in daily life. Hyperscanning EEG is the most appropriate way to investigate this, because of its portability, low cost, and the fact that the device is easy to wear (Stopczynski et al., 2014). This simplicity enables us to acquire data about brain activation even in extremely intricate social situations without burden (Gevins et al., 2012). Recently, several hyperscanning EEG studies revealed inter-brain effects in social interactions such as playing music with others (Lindenberger et al., 2009; Sänger et al., 2012, 2013; Müller et al., 2013), playing games (Astolfi et al., 2010; De Vico Fallani et al., 2010), and verbal communication (Kawasaki et al., 2013). These results raise the expectation that inter-brain EEG effects could represent effective neuromarkers for the quality of social communication in our daily life.

However, EEG has a limited ability to define brain regions exhibiting significant inter-brain effects due to its low spatial resolution (Kaiboriboon et al., 2012) and ill-posed inverse problem (Grech et al., 2008; Hari et al., 2013). We propose to use hyperscanning EEG–fMRI to overcome the limitations of EEG. Simultaneous EEG–fMRI has been used to achieve high temporal and spatial resolution in studies of human brain function. Previous studies have proposed several analytical methods for combining EEG and fMRI data. The first approach is the fMRI-informed EEG method, which aims to guide electromagnetic source estimation using results obtained from fMRI images with superior spatial resolution, thereby improving EEG source localization (Heinze et al., 1994; Huster et al., 2011). The second method for handling multimodal neuroimaging data is EEG-informed fMRI, which considers the direct correlation between EEG and fMRI. In this method, we first could detect specific EEG features of interest, such as ERP amplitude (Debener et al., 2005), EEG synchronization (Mizuhara et al., 2005), and power within specific frequency bands (Laufs et al., 2003; Scheeringa et al., 2009) over time. Next, we investigate the association between fluctuations of these features over time, as well as fMRI signal fluctuation during the experiment. The third method is joint independent component analysis (ICA) (Calhoun et al., 2006; Huster et al., 2012). In this method, a common or symmetric model is used to jointly assess information from both EEG and fMRI data. In joint ICA, each type of neuroimaging data (i.e., EEG and fMRI data) is preprocessed independently. Subsequently, to examine the relationship between EEG and fMRI data, fMRI statistical maps and ERP data of all subjects are merged into a single matrix and subjected to joint ICA. This approach provides joint spatiotemporal decomposition with joint independent components corresponding to electrophysiologically measured responses associated with clusters of activated regions (Calhoun et al., 2006; Huster et al., 2012). By adopting these methods, even in hyperscanning studies, we could combine high temporal resolution EEG data with high-spatial resolution fMRI data, and precisely define the neural basis of inter-brain EEG effects.

If we could precisely define the neural basis of inter-brain EEG effects, then it would be perfectly acceptable to use these effects as neuromarkers of social interaction. There are two ways to use inter-brain EEG effects as neuromarkers. The first is as a neuromarker of the quality of social interactions in daily life. For example, in practice, an inter-brain effect could be a measure of successful learning processes in educational environments. Learning processes, especially in school, involve social interactions between teacher and students (Watanabe, 2013; Verga and Kotz, 2013), and on-line interaction is essential for these processes (Kuhl et al., 2003). Such interactions consist of two individuals, and cannot be reduced to the individual level (Hari and Kujala, 2009; Konvalinka and Roepstorff, 2012). Therefore, we hope that we could use inter-brain effects as neuromarkers of the quality of the learning process. So far, no study has directly investigated whether inter-brain effects reflect the quality of learning processes. However, performance of the JA task, which is proposed to be at the core of the evolution of verbal language (Pinker, 2010), is related to emergence of inter-brain neural synchronization in the right IFG (Saito et al., 2010; Tanabe et al., 2012). In addition, verbal communication evokes inter-brain EEG coherence on the left IFG (Jiang et al., 2012). These lines of evidence suggest that the quality of communication during the learning process might be reflected in inter-brain effects in the IFG.

In our social interactions, such as learning processes, the relationship between two individuals is asymmetric, and each individual plays different role (e.g., speaker and listener). When there is asymmetry of roles in communication, causality between two brains might be a better neuromarker of interaction quality than coherence or synchronization. For example, during verbal communication, causality of EEG between team members in a cooperative relationship is more significant than causality between individuals on different teams, who are in a competitive relationship (Astolfi et al., 2010). Thus, causality could be a better neuromarker of the quality of asymmetric social interaction. For example, causality from teacher's brain to student's brain could be used as a neuromarker of how effectively the teacher can make the student understand. To this end, hyperscanning EEG–fMRI is valuable in precisely defining brain regions associated with EEG causality in asymmetrical social interactions. Causality between two brains should be investigated more intensively in future studies.

Not only the quality of social interaction, but also our behavior and psychological perspectives on social interactions, could be quantified using inter-brain effects. Individual behavior and psychological perspectives can be understood in the individual brain (Lahnakoski et al., 2014). However, our behavior and psychological perspectives during social interactions are clearly influenced by those of our partners. Therefore, inter-brain effects could provide better markers of behavior and psychological perspective than activations within a single brain. As noted above, a subject's intention to defect on their partner could be predicted with high accuracy before the decision by evaluating changes in the inter-brain network pattern (De Vico Fallani et al., 2010). In our future studies, we have to test whether the neuromarker from two communicating brains is better than the neuromarker from a single brain.

6. Necessity of the mathematical model

It is well known that multiple organisms can constitute a single dynamic complex system through their interactions, such as the synchronized flickering of certain types of fireflies (Buck, 1988) and the synchronized cries of Japanese tree frogs (Aihara et al., 2014). For these phenomena, mathematical modeling has played an important role in revealing the mechanism underlying the emergence of synchronization, as well as its function (Ermentrout, 1991; Aihara et al., 2014). As with these phenomena, two individuals who are coupled through social interaction constitute a dynamic complex system that cannot be reduced to the individual level (Beer, 2000; Hari and Kujala, 2009; Froese et al., 2013). Therefore, a mathematical model would help us to understand the nature of inter-brain effects that reflect properties of social interaction. Modeling studies and hyperscanning experiments could contribute, in a complementary manner, to our understanding of the nature of inter-brain neural synchronization. First, we could build a mathematical model based on the results of an experiment. Next, we could confirm whether the model's predictions accurately explain actual phenomena, and then revise the model. By repeating this model validation cycle, we could get asymptotically closer to the nature of two-in-one systems consisting of two individuals.

A few modeling studies have addressed the mechanisms underlying emergence of inter-brain neural effects. Dumas et al. (2012)

investigated how anatomical connectivity influenced the emergence of inter-brain EEG synchronization as subjects imitated a partner's hand gestures (Dumas et al., 2010). In that study, the authors designed whole-brain numerical simulations that combined a real connectome dataset (Iturria-Medina et al., 2007) with a Kuramoto model (Kuramoto, 1975); in particular, they made the model simulate brain behavior in the resting state. After the validation, they added virtual sensorimotor coupling by linking the motor area of each virtual brain with the visual regions of the other (Hari and Kujala, 2009), in order to evoke inter-brain neural synchronization between the two virtual brains. After that, using the model, they investigated the roles played by anatomical structures in the emergence of inter-brain synchronization via the perception–action loop. First, the model revealed that similarity of anatomical structure between two brains makes brain activation dynamics more similar, even when there is no interaction between them. In addition, when the strength of intra-brain connectivity increased, inter-brain synchronization was enhanced to a greater extent by real connectome data than by random connection data. These results support the idea that the anatomical structure of our brain plays a key role in the emergence of inter-brain neural synchronization (Dumas et al., 2012).

By comparing the prediction of our model with actual phenomena, we can validate the model. The model predicts that similarity of anatomical structure between two brains results in weak inter-brain neural synchronization. Differences in the network organization of ASD patients from that of normal control subjects have been frequently observed using fMRI and diffusion tensor imaging techniques (Anagnostou and Taylor, 2011). In addition, the inter-brain correlation analysis suggested that the increased randomness of regional brain activity in high functioning ASD patient (Salmi et al., 2013). These lines of evidence suggest that the inter-brain synchronization between normal control and ASD participants should be weak because of the low degree of similarity of anatomical connectivity. Consistent with this, only weak inter-brain synchronization on the right IFG is observed between normal controls and ASD participants (Tanabe et al., 2012). The agreement between model prediction and observed synchronization suggests that the assumption of the mathematical model is valid (Dumas et al., 2012).

Another important advantage of mathematical models is that they can make new predictions about unknown phenomena. For example, the model of Dumas et al. (2012) predicts that training could influence the emergence of inter-brain neural effects, because it is well known that training can change the properties of intra-brain networks (for review, Guerra-Carrillo et al., 2014). This prediction seems to be reasonable, because our social skills are acquired through communication with other individuals, starting at birth and continuing throughout our lives (Johnson et al., 2005; Grossmann and Johnson, 2007; Murray and Trevarthen, 1985; Striano et al., 2006). Therefore, if the emergence of inter-brain neural synchronization reflects our social skills, the experience of social interaction might increase or decrease this synchronization. We can test this idea by investigating whether training has an effect on the properties of synchronization. To the best of our knowledge, few studies have demonstrated that inter-brain synchronization of the EEG waveform can be enhanced by means of training (Yun et al., 2012; Kawasaki et al., 2013). If training has a direct effect on the emergence of inter-brain neural synchronization through social Hebbian learning (Wolpert et al., 2003; Keysers and Perrett, 2004), then the region related to training region might overlap with the region involved in synchronization. To date, no study has clearly demonstrated such an overlap (Yun et al., 2012; Kawasaki et al., 2013), in part because EEG has very limited power to localize the center of activation (Hari et al., 2013). In future studies, by combining inter-brain EEG synchronization with inter-brain fMRI synchronization, we could test the effect of training on the emergence of inter-brain effects, as predicted by the mathematical model.

Hyperscanning EEG–fMRI allows us to look at the two-in-one system from a different perspective. Consequently, in future studies, evidence obtained in hyperscanning EEG–fMRI experiments might play a major role in constructing a mathematical model of inter-brain effects in social interactions. In hyperscanning EEG studies, inter-brain effects exhibit a significant frequency dependency (Dumas et al., 2010; Astolfi et al., 2010; Kawasaki et al., 2013). The EEG in distinct frequency bands is assumed to reflect different aspects of our cognitive process (Scheeringa et al., 2011). Therefore, when we use hyperscanning EEG–fMRI and joint ICA, fMRI data can precisely define the intra- and inter-brain network that is associated with the inter-brain EEG network in different frequency bands. fMRI networks associated with different EEG frequencies might reflect different aspects of social interaction. In addition, joint ICA could be used to analyze any type of data. Therefore, hyperscanning EEG–fMRI in conjunction with joint ICA could allow us to analyze multimodal neuroimaging data and behavioral data via a unified method. In social interactions, the two-in-one system arises via the behavior–perception loop (Hari and Kujala, 2009; Froese et al., 2013). Hyperscanning EEG–fMRI allows us to analyze almost all aspects of social interactions at high spatial and temporal resolution. The evidence obtained could contribute to construction of mathematical models that could account for the nature of inter-brain effects in social interactions.

7. Concluding remarks

Here, we reviewed studies of inter-brain neural coupling during social interactions that used hyperscanning neuroimaging techniques, i.e. EEG, fNIRS and fMRI. Although these hyperscanning studies have revealed inter-brain effects in social interactions, the limits of neuroimaging techniques place limitations on the information that can be obtained in these studies. Hyperscanning fMRI can precisely detect the regions exhibiting inter-brain effects; however, it is impossible to record such effects during daily life. The characteristics of hyperscanning EEG are in contrast to those of hyperscanning fMRI. The portability of EEG devices allows us to record inter-brain effect in very natural setting; however, the ability of EEG to localize the epicenter of brain activation is very limited. The characteristics of hyperscanning fNIRS are intermediate between those of hyperscanning EEG and fMRI. We propose the use of hyperscanning EEG–fMRI as a means of understanding the nature of inter-brain effects at high spatial and temporal resolution. This method will provide an opportunity to use inter-brain effects as neuromarkers of social interactions in daily life. In addition, mathematical modeling of inter-brain effects will play important roles in future studies. We think that the combination of hyperscanning EEG–fMRI and mathematical modeling could make a dramatic contribution to our understanding of on-line social interactions.

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References

Aihara, I., Mizumoto, T., Otsuka, T., Awano, H., Nagira, K., Okuno, H.G., Aihara, K., 2014. Spatio-temporal dynamics in collective frog choruses examined by mathematical modeling and field observations. Sci. Rep. 4, 3891.

Allison, T., Puce, a, McCarthy, G., 2000. Social perception from visual cues: role of the STS region. Trends Cogn. Sci. 4 (7), 267–278.

Anagnostou, E., Taylor, M.J., 2011. Review of neuroimaging in autism spectrum disorders: what have we learned and where we go from here. Mol. Autism 2 (1), 4.

Astolfi, L., Toppi, J., De Vico Fallani, F., Vecchiato, G., Cincotti, F., Wilke, C.T., Yuan, H., Mattia, D., Salinari, S., He, B., Babiloni, F., 2011. Imaging the social brain by simultaneous hyperscanning during subject interaction. IEEE Intell. Syst. 26 (5), 38–45.

Astolfi, L., Toppi, J., De Vico Fallani, F., Vecchiato, G., Salinari, S., Mattia, D., Cincotti, F., Babiloni, F., 2010. Neuroelectrical hyperscanning measures simultaneous brain activity in humans. Brain Topogr. 23 (3), 243–256.

Babiloni, F., Astolfi, L., 2012. Social neuroscience and hyperscanning techniques: past, present and future. Neurosci. Biobehav. Rev., 1–36.

Baess, P., Zhdanov, A., Mandel, A., Parkkonen, L., Hirvenkari, L., Mäkelä, J.P., Jousmäki, V., Hari, R., 2012. MEG dual scanning: a procedure to study real-time auditory interaction between two persons. Front. Hum. Neurosci. 6, 83.

Beer, R., 2000. Dynamical approaches to cognitive science. Trends Cogn. Sci. 4 (3), 91–99.

Buck, J., 1988. Synchronous rhythmic flashing in fireflys. II. Q. Rev. Biol. 63, 265–289.

Burgess, A.P., 2013. On the interpretation of synchronization in EEG hyperscanning studies: a cautionary note. Front. Hum. Neurosci. 7, 881.

Calhoun, V.D., Adali, T., Pearlson, G.D., Kiehl, K.A., 2006. Neuronal chronometry of target detection: fusion of hemodynamic and event-related potential data. Neuroimage 30 (2), 544–553.

Cordes, D., Haughton, V.M., Arfanakis, K., Carew, J.D., Turski, P.A., Moritz, C.H., Quigley, M.A., Meyerand, M.E., 2001. Frequencies contributing to functional connectivity in the cerebral cortex in "resting-state" data. Am. J. Neuroradiol. 22 (7), 1326–1333.

Cui, X., Bryant, D.M., Reiss, A.L., 2012. NIRS-based hyperscanning reveals increased interpersonal coherence in superior frontal cortex during cooperation. Neuroimage 59 (3), 2430–2437.

Debener, S., Ullsperger, M., Siegel, M., Fiehler, K., von Cramon, D.Y., Engel, A.K., 2005. Trial-by-trial coupling of concurrent electroencephalogram and functional magnetic resonance imaging identifies the dynamics of performance monitoring. J. Neurosci. 25 (50), 11730–11737.

De Vico Fallani, F., Nicosia, V., Sinatra, R., Astolfi, L., Cincotti, F., Mattia, D., Wilke, C., Doud, A., Latora, V., He, B., Babiloni, F., 2010. Defecting or not defecting: how to "read" human behavior during cooperative games by EEG measurements. PLoS ONE 5 (12), e14187.

Dommer, L., Jäger, N., Scholkmann, F., Wolf, M., Holper, L., 2012. Between-brain coherence during joint n-back task performance: a two-person functional near-infrared spectroscopy study. Behav. Brain Res. 234 (2), 212–222.

Duane, A.T.D., Behrendt, T., 1965. Extrasensory electroencephalographic induction between identical twins. Science 150 (3964), 367.

Dumas, G., Nadel, J., Soussignan, R., Martinerie, J., Garnero, L., 2010. Inter-brain synchronization during social interaction. PLoS ONE 5 (8), e12166.

Dumas, G., 2011. Towards a two-body neuroscience. Commun. Integr. Biol. 4 (3), 349–352.

Dumas, G., Chavez, M., Nadel, J., Martinerie, J., 2012. Anatomical connectivity influences both intra- and inter-brain synchronizations. PLoS ONE 7 (5), e36414.

Englander, Z.A., Haidt, J., Morris, J.P., 2012. Neural basis of moral elevation demonstrated through inter-subject synchronization of cortical activity during free-viewing. PLoS ONE 7 (6), e39384.

Ermentrout, B., 1991. An adaptive model for synchrony in the firfly Pteroptyx malaccae. J. Math. Biol. 29, 571–585.

Fair, D., Schlaggar, B., Cohen, A., Miezin, F., Dosenbach, N.U.F., Wenger, K.K., Fox, M.D., Snyder, A.Z., Raichle, M.E., Petersen, S.E., 2007. A method for using blocked and event-related fMRI data to study "resting state" functional connectivity. Neuroimage 35 (1), 396–405.

Fliessbach, K., Phillipps, C.B., Trautner, P., Schnabel, M., Elger, C.E., Falk, A., Weber, B., 2012. Neural responses to advantageous and disadvantageous inequity. Front. Hum. Neurosci., 165.

Frith, C., Frith, U., 2001. The biological basis of social interaction. Curr. Dir. Psychol. Sci. 10, 51–155.

Froese, T., Iizuka, H., Ikegami, T., 2013. From synthetic modeling of social interaction to dynamic theories of brain-body-environment-body-brain systems. Behav. Brain Sci. 36 (4), 420–421.

Funane, T., Kiguchi, M., Atsumori, H., Sato, H., Kubota, K., Koizumi, H., 2011. Synchronous activity of two people's prefrontal cortices during a cooperative task measured by simultaneous near-infrared spectroscopy. J. Biomed. Opt. 16 (7), 077011.

Gallagher, H.L., Frith, C.D., 2003. Functional imaging of "theory of mind". Trends Cogn. Sci. 7 (2), 77–83.

Gallotti, M., Frith, C.D., 2013. Social cognition in the we-mode. Trends Cogn. Sci. 17 (4), 160–165.

Guerra-Carrillo, B., Mackey, A.P., Bunge, S.A., 2014. Resting-state fMRI: a window into human brain plasticity. Neuroscientist 20 (5), 522–533.

Gevins, A., Chan, C.S., Sam-Vargas, L., 2012. Towards measuring brain function on groups of people in the real world. PLoS ONE 7 (9), e44676.

Grech, R., Cassar, T., Muscat, J., Camilleri, K.P., Fabri, S.G., Zervakis, M., Xanthopoulos, P., Sakkalis, V., Vanrumste, B., 2008. Review on solving the inverse problem in EEG source analysis. J. Neuroeng. Rehabil. 5, 25.

Gregg, N.M., White, B.R., Zeff, B.W., Berger, A.J., Culver, J.P., 2010. Brain specificity of diffuse optical imaging: improvements from superficial signal regression and tomography. Front. Neuroenerg. 2, 14.

Grossmann, T., Johnson, M.H., 2007. The development of the social brain in human infancy. Eur. J. Neurosci. 25 (4), 909–919.

Hari, R., Kujala, M.V., 2009. Brain basis of human social interaction: from concepts to brain imaging. Physiol. Rev. 89, 453–479.

Hari, R., Himberg, T., Nummenmaa, L., Hämäläinen, M., Parkkonen, L., 2013. Synchrony of brains and bodies during implicit interpersonal interaction. Trends Cogn. Sci. 17 (3), 105–106.

Hasson, U., Nir, Y., Levy, I., Fuhrmann, G., Malach, R., 2004. Intersubject synchronization of cortical activity during natural vision. Science 303 (5664), 1634–1640.

Hasson, U., Ghazanfar, A.A., Galantucci, B., Garrod, S., Keysers, C., 2012. Brain-to-brain coupling: a mechanism for creating and sharing a social world. Trends Cogn. Sci. 16 (2), 114–121.

Heinze, H.J., Mangun, G.R., Burchert, W., Hinrichs, H., Scholz, M., Munte, T.F., Gös, A., Scerg, M., Johannes, S., Hundeshagen, H., Gazzaniga, M.S., Hillyard, S.A., 1994. Combined spatial and temporal imaging of brain activity during visual selective attention in humans. Nature 372 (8), 543–546.

Holper, L., Scholkmann, F., Wolf, M., 2012. Between-brain connectivity during imitation measured by fNIRS. Neuroimage 63 (1), 212–222.

Huettel, S.A., Song, A.W., McCarthy, G., 2009. Functional Magnetic Resonance Imaging, second ed. Sinauer, Sunderland, MA.

Huster, R.J., Eichele, T., Enriquez-Geppert, S., Wollbrink, A., Kugel, H., Konrad, C., Pantev, C., 2011. Multimodal imaging of functional networks and event-related potentials in performance monitoring. Neuroimage 56 (3), 1588–1597.

Huster, R.J., Debener, S., Eichele, T., Herrmann, C.S., 2012. Methods for simultaneous EEG–fMRI: an introductory review. J. Neurosci. 32 (18), 6053–6060.

Iturria-Medina, Y., Canales-Rodríguez, E.J., Melie-García, L., Valdés-Hernández, P.A., Martínez-Montes, E., Alemán-Gómez, Y., Sánchez-Bornot, J.M., 2007. Characterizing brain anatomical connections using diffusion weighted MRI and graph theory. Neuroimage 36 (3), 645–660.

Jiang, J., Dai, B., Peng, D., Zhu, C., Liu, L., Lu, C., 2012. Neural synchronization during face-to-face communication. J. Neurosci. 32 (45), 16064–16069.

Johnson, M.H., Griffin, R., Csibra, G., Halit, H., Farroni, T., de Haan, M., Tucker, L.A., Baron-Cohen, S., Richards, J., 2005. The emergence of the social brain network: evidence from typical and atypical development. Dev. Psychopathol. 17, 599–619.

Kaiboriboon, K., Lüders, H.O., Hamaneh, M., Turnbull, J., Lhatoo, S.D., 2012. EEG source imaging in epilepsy – practicalities and pitfalls. Nat. Rev. Neurol. 8, 498–507.

Kawasaki, M., Yamada, Y., Ushiku, Y., Miyauchi, E., Yamaguchi, Y., 2013. Inter-brain synchronization during coordination of speech rhythm in human-to-human social interaction. Sci. Rep. 3, 1692.

Keysers, C., Perrett, D.I., 2004. Demystifying social cognition: a Hebbian perspective. Trends Cogn. Sci. 8 (11), 501–507.

King-Casas, B., Tomlin, D., Anen, C., Camerer, C.F., Quartz, S.R., Montague, P.R., 2005. Getting to know you: reputation and trust in a two-person economic exchange. Science 308 (5718), 78–83.

Kirilina, E., Jelzow, A., Heine, A., Niessing, M., Wabnitz, H., Brühl, R., Ittermann, B., Jacobs, A.M., Tachtsidis, I., 2012. The physiological origin of task-evoked systemic artefacts in functional near infrared spectroscopy. Neuroimage 61 (1), 70–81.

Konvalinka, I., Roepstorff, A., 2012. The two-brain approach: how can mutually interacting brains teach us something about social interaction? Front. Hum. Neurosci. 6, 215.

Krill, A.L., Platek, S.M., 2012. Working together may be better: activation of reward centers during a cooperative maze task. PLoS ONE 7 (2), e30613.

Kuhl, P.K., Tsao, F.-M., Liu, H.-M., 2003. Foreign-language experience in infancy: effects of short-term exposure and social interaction on phonetic learning. Proc. Natl. Acad. Sci. U. S. A. 100 (15), 9096–9101.

Kuramoto, Y., 1975. In: Araki, H. (Ed.), Self-entrainment of Population of Coupled Non-linear Oscillators, vol. 39. Springer-Verlag, Berlin, Heidelberg, pp. 420–422.

Lahnakoski, J.M., Glerean, E., Jääskeläinen, I.P., Hyönä, J., Hari, R., Sams, M., Nummenmaa, L., 2014. Synchronous brain activity across individuals underlies shared psychological perspectives. Neuroimage 100 (15), 316–324.

Laufs, H., Kleinschmidt, A., Beyerle, A., Eger, E., Salek-Haddadi, A., Preibisch, C., Krakow, K., 2003. EEG-correlated fMRI of human alpha activity. Neuroimage 19 (4), 1463–1476.

Lindenberger, U., Li, S.-C., Gruber, W., Müller, V., 2009. Brains swinging in concert: cortical phase synchronization while playing guitar. BMC Neurosci. 10, 22.

Liu, T., Pelowski, M., 2014. Clarifying the interaction types in two-person neuroscience research. Front. Hum. Neurosci. 8, 276.

Mizuhara, H., Wang, L.Q., Kobayashi, K., Yamaguchi, Y., 2005. Long-range EEG phase synchronization during an arithmetic task indexes a coherent cortical network simultaneously measured by fMRI. Neuroimage 27 (3), 553–563.

Moeller, S., Yacoub, E., Olman, C.A., Auerbach, E., Strupp, J., Harel, N., Uğurbil, K., 2010. Multiband multislice GE-EPI at 7 tesla, with 16-fold acceleration using partial parallel imaging with application to high spatial and temporal whole-brain fMRI. Magn. Reson. Med. 63 (5), 1144–1153.

Montague, P.R., Berns, G.S., Cohen, J.D., McClure, S.M., Pagnoni, G., Dhamala, M., Wiest, M.C., Karpov, I., King, R.D., Apple, N., Fisher, R., 2002. Hyperscanning: simultaneous fMRI during linked social interactions. Neuroimage 16 (4), 1159–1164.

Morita, T., Tanabe, H.C., Sasaki, A.T., Shimada, K., Kakigi, R., Sadato, N., 2014. The anterior insular and anterior cingulate cortices in emotional processing for self-face recognition. Soc. Cogn. Affect. Neurosci. 9 (5), 570–579.
Murray, L., Trevarthen, C., 1985. Emotional regulations of interactions between two-month-olds and their mothers. In: Field, T.M., Fox, N.A. (Eds.), Social Perception in Infants. Alex, Norwood, pp. 177–197.
Müller, V., Sänger, J., Lindenberger, U., 2013. Intra- and inter-brain synchronization during musical improvisation on the guitar. PLOS ONE 8 (9), e73852.
Naeem, M., Prasad, G., Watson, D.R., Kelso, J.A.S., 2012. Electrophysiological signatures of intentional social coordination in the 10–12 Hz range. Neuroimage 59 (2), 1795–1803.
Osaka, N., Minamoto, T., Yaoi, K., Azuma, M., Osaka, M., 2014. Neural synchronization during cooperated humming: a hyperscanning study using fNIRS. Proc. Soc. Behav. Sci. 126, 241–243.
Pascual-Marqui, R.D., Michel, C.M., Lehmann, D., 1994. Low resolution electromagnetic tomography: a new method for localizing electrical activity in the brain. Int. J. Psychophysiol. 18, 49–65.
Pinker, S., 2010. Colloquium paper: the cognitive niche: coevolution of intelligence, sociality, and language. Proc. Natl. Acad. Sci. U. S. A. 107 (Suppl. 2), 8993–8999.
Saito, D.N., Tanabe, H.C., Izuma, K., Hayashi, M.J., Morito, Y., Komeda, H., Uchiyama, H., Kosaka, H., Okazawa, H., Fujibayashi, Y., Sadato, N., 2010. "Stay tuned": inter-individual neural synchronization during mutual gaze and joint attention. Front. Integr. Neurosci. 4, 127.
Salmi, J., Roine, U., Glerean, E., Lahnakoski, J., Nieminen-von Wendt, T., Tani, P., Leppämäki, S., Nummenmaa, L., Jääskeläinen, I.P., Carlson, S., Rintahaka, P., Sams, M., 2013. The brains of high functioning autistic individuals do not synchronize with those of others. Neuroimage Clin. 3, 489–497.
Sänger, J., Lindenberger, U., Müller, V., 2011. Interactive brains, social minds. Commun. Integr. Biol. 4 (6), 655–663.
Sänger, J., Müller, V., Lindenberger, U., 2012. Intra- and inter-brain synchronization and network properties when playing guitar in duets. Front. Hum. Neurosci. 6, 312.
Sänger, J., Müller, V., Lindenberger, U., 2013. Directionality in hyperbrain networks discriminates between leaders and followers in guitar duets. Front. Hum. Neurosci. 7, 234.
Scheeringa, R., Petersson, K.M., Oostenveld, R., Norris, D.G., Hagoort, P., Bastiaansen, M.C., 2009. Trial-by-trial coupling between EEG and BOLD identifies networks related to alpha and theta EEG power increases during working memory maintenance. Neuroimage 44 (3), 1224–1238.
Scheeringa, R., Fries, P., Petersson, K.M., Oostenveld, R., Grothe, I., Norris, D.G., Hagoort, P., Bastiaansen, M.C., 2011. Neuronal dynamics underlying high- and low-frequency EEG oscillations contribute independently to the human BOLD signal. Neuron 69 (3), 572–583.
Schilbach, L., Timmermans, B., Reddy, V., Costall, A., Bente, G., Schlicht, T., Vogeley, K., 2013. Toward a second-person neuroscience. Behav. Brain Sci. 36 (4), 393–414.
Scholkmann, F., Holper, L., Wolf, U., Wolf, M., 2013. A new methodical approach in neuroscience: assessing inter-personal brain coupling using functional near-infrared imaging (fNIRI) hyperscanning. Front. Hum. Neurosci. 7, 813.
Smith, A., Lohrenz, T., King, J., Montague, P.R., Camerer, C.F., 2014. Irrational exuberance and neural crash warning signals during endogenous experimental market bubbles. Proc. Natl. Acad. Sci. U. S. A. 111 (29), 10503–10508.
Spiegelhalder, K., Ohlendorf, S., Regen, W., Feige, B., Tebartz van Elst, L., Weiller, C., Henning, J., Berger, M., Tüscher, O., 2014. Interindividual synchronization of brain activity during live verbal communication. Behav. Brain Res. 258, 75–79.
Sporns, O., Zwi, J.D., 2004. The small world of the cerebral cortex. Neuroinfomatics 2 (2), 145–162.
Stopczynski, A., Stahlhut, C., Larsen, J.E., Petersen, M.K., Hansen, L.K., 2014. The smartphone brain scanner: a portable real-time neuroimaging system. PLOS ONE 9 (2), e86733.
Striano, T., Henning, A., Stahl, D., 2006. Sensitivity to interpersonal timing at 3 and 6 months of age. Interact. Stud. 7 (2), 251–271.
Tanabe, H.C., Kosaka, H., Saito, D.N., Koike, T., Hayashi, M.J., Izuma, K., Komeda, H., Ishitobi, M., Omori, M., Munesue, T., Okazawa, H., Wada, Y., Sadato, N., 2012. Hard to "tune in": neural mechanisms of live face-to-face interaction with high-functioning autistic spectrum disorder. Front. Hum. Neurosci. 6, 268.
Tognoli, E., Lagarde, J., DeGuzman, G.C., Kelso, J.A.S., 2007. The phi complex as a neuromarker of human social coordination. Proc. Natl. Acad. Sci. U. S. A. 104 (19), 8190–8195.
Verga, L., Kotz, S.A., 2013. How relevant is social interaction in second language learning? Front. Hum. Neurosci. 7, 550.
Watanabe, K., 2013. Teaching as a dynamic phenomenon with interpersonal interactions. Mind Brain Educ. 7 (2), 91–100.
Wolpert, D.M., Doya, K., Kawato, M., 2003. A unifying computational framework for motor control and social interaction. Philos. Trans. R. Soc. Lond. Ser. B: Biol. Sci. 358 (1431), 593–602.
Wood, J.N., Grafman, J., 2003. Human prefrontal cortex: processing and representational perspectives. Nat. Rev. Neurosci. 4 (2), 139–147.
Yun, K., Watanabe, K., Shimojo, S., 2012. Interpersonal body and neural synchronization as a marker of implicit social interaction. Sci. Rep. 2, 959.